Suitable for Linked List & it's k passes are not useful.

2.) INSERTION SORT -

Min^m. - $O(1)$
Max^m. - $O(n)$ } This is for inserting an element.

Ex.

A

8	5	7	3	2

Step 1 Assume element 8 is sorted & compare elements to it & then swap.

I^st pass

5	8	7	3	2

II^nd Pass

5	7	8	3	2

III^rd Pass

3	5	7	8	2

IV^th Pass

2	3	5	7	8

No. of passes = $n-1$

No. of comparisons = $\frac{(n-1)n}{2} = O(n^2)$

Max no. of swaps = $\frac{n(n-1)}{2} = O(n^2)$

It is more useful & compatible in linked list than array.

It is adaptive in nature.

Min^m. time Complexity - $O(n)$

Max^m. Time Complexity - $O(n^2)$

It is stable

No, extra space consumed also.

Advantages - • Easy to implement & efficient to use on small sets of data.

• It requires less memory space. • It can be efficiently implemented on data sets that are substantially sorted.